Sabancı University

Information Visualization Homework Report

Population Change and Socio-Economic Indicators in Türkiye (2019–2024)

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Project Question

What relationship exists between population growth in major Turkish cities and national socio-economic indicators such as unemployment rate, GDP per capita, and inflation between 2019 and 2024?

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1 Introduction

This report presents the design process and implementation for a visualization homework project based on Turkish population and socio-economic data. The study focuses on the years 2019 to 2024 and investigates whether changes in the populations of selected major cities move together with national indicators such as unemployment rate, GDP per capita, and inflation.

The report combines hand-drawn sketches, design rationale, and one coded visualization. Some sections intentionally contain placeholders so that additional handwritten images, screenshots, reflections, and course-specific terminology can be inserted later.

Main research question:

What relationship exists between population growth in major Turkish cities and national socio-economic indicators such as unemployment rate, GDP per capita, and inflation between 2019 and 2024?

2 Dataset Description

2.1 Scope of the Study

Although the original census dataset covered a broader time interval, only the years between 2019 and 2024 were used in this homework in order to align the population data with the socio-economic indicators collected for the same period.

The selected cities are:

- Van
- Muğla
- Adana
- Ankara
- İstanbul
- İzmir

2.2 Population Data Used

Table 1: Population of Selected Turkish Cities (2019–2024)

Year	Van	Muğla	Adana	Ankara	İstanbul	İzmir
2019	1,136,757	983,142	2,237,940	5,639,076	15,519,267	4,367,251
2020	1,149,342	1,000,773	2,258,718	5,663,322	15,462,452	4,394,694
2021	1,141,015	1,021,141	2,263,373	5,747,325	15,840,900	4,425,789
2022	1,128,749	1,048,185	2,274,106	5,782,285	15,907,951	4,462,056
2023	1,127,612	1,066,736	2,270,298	5,803,482	15,655,924	4,479,525
2024	1,118,087	1,081,867	2,280,484	5,864,049	15,701,602	4,493,242

2.3 Average Population Values Used in Some Sketches

For some sketches, an average city population value was used for simplicity and readability.

Table 2: Average Population Used in Simplified Sketches

Year	Average Population
2019	4,980,572
2020	5,021,550
2021	5,073,257
2022	5,100,555
2023	5,067,263
2024	5,089,889

2.4 Socio-Economic Indicators from TÜİK

The three national indicators selected from TÜİK for the same period are unemployment rate, GDP per capita (USD), and inflation rate.

Table 3: National Socio-Economic Indicators of Türkiye (2019–2024)

Year	Unemployment Rate	GDP per Capita (USD)	Inflation Rate
2019	13.7	9127	11.84
2020	13.2	8555	14.60
2021	12.0	9592	36.08
2022	10.4	10655	62.93
2023	9.4	13243	64.00
2024	8.6	15325	44.38

3 Task 1: Hand-Drawn Exploratory Sketches

3.1 Task 1A: Türkiye Map with City-Based Time Series

In this task, a hand-drawn map of Türkiye was created, and the six selected cities were placed and labeled on the map. Next to each city, a small time-series line chart was drawn to show population change between 2019 and 2024.

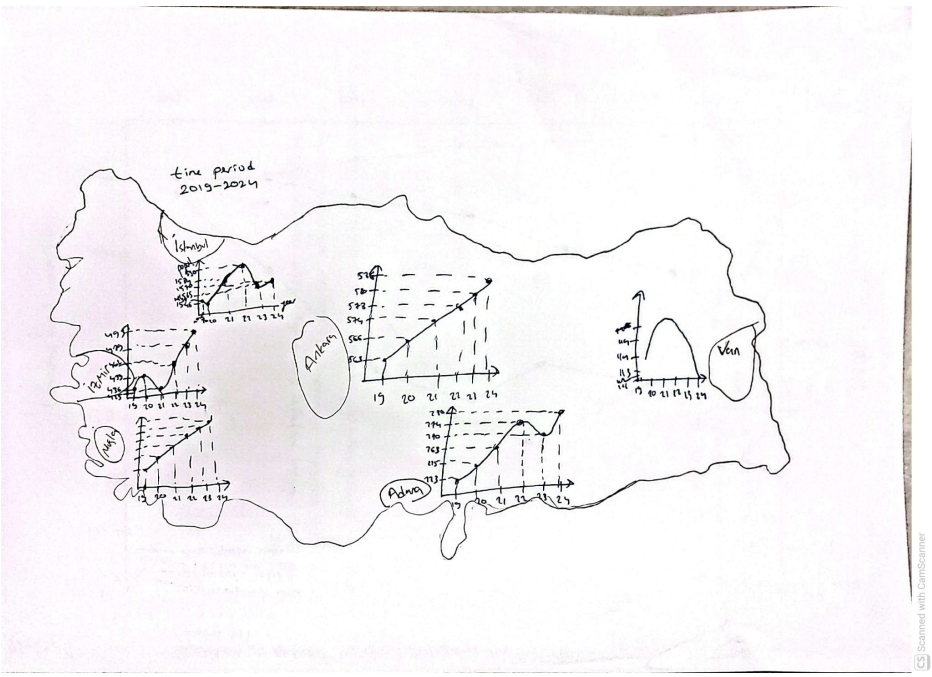

Figure 1: Hand-drawn Türkiye map with labeled cities and mini line charts (Task 1).

3.2 Task 1B: Three Socio-Economic Indicators as Line Charts

In this task, the three national indicators were drawn as separate line charts on paper for the same time period (Unfortunately, the image did not fit on this page; please see the next page.).

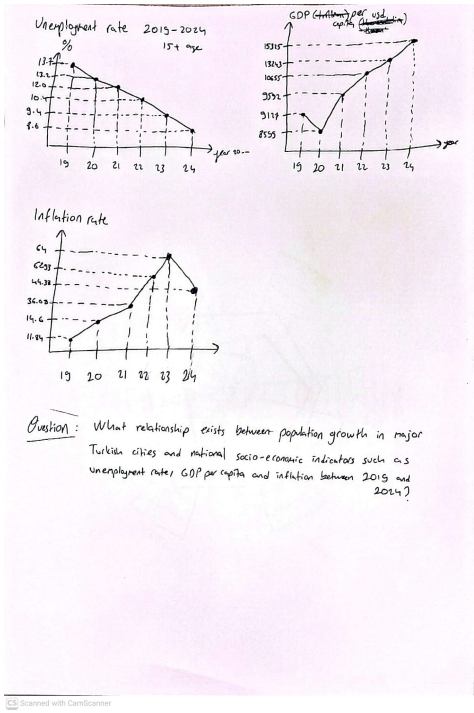

Figure 2: Hand-drawn line charts of unemployment rate, GDP per capita, and inflation (Task 1B).

Short explanation placeholder:

Write a short explanation of why line charts were chosen, what temporal trends are visible, and how these charts support later comparison.

3.3 Task 1C: Formulated Question

Based on Task 1A and Task 1B, the following question was formulated:

> **What relationship exists between population growth in major Turkish cities and national socio-economic indicators such as unemployment rate, GDP per capita, and inflation between 2019 and 2024?**

Why this question?

Write 1–2 paragraphs explaining why this question naturally emerges from the data and why visualization is needed to answer it effectively.

4 Task 2: Three Visualization Sketches

This section presents three visualization sketches designed to answer the main question. One of the sketches is intentionally more abstract, as requested in the homework. For each sketch, placeholders are provided for the image and explanation.

4.1 Sketch 1: Multi-Line Comparison View

Figure 3: Placeholder for Sketch 1 image. Insert the hand-drawn or digital sketch here.

Visualization type: *Example: multi-line chart / coordinated line chart / small multiples*

Goal: *Explain what this sketch tries to reveal.*

Design rationale:

Explain what variables are encoded using position, color, slope, or grouping. Mention why this representation is appropriate for temporal comparison. Use class terminology such as marks, channels, encoding, scale, comparison, trend, overview, or correlation.

Where this could be useful:

Write where or for whom this visualization would be useful.

Strengths:

- *Strength 1*
- *Strength 2*
- *Strength 3*

Weaknesses:

- *Weakness 1*
- *Weakness 2*

4.2 Sketch 2: Map-Based or Small-Multiple Analytical View

Figure 4: Placeholder for Sketch 2 image. Insert the hand-drawn or digital sketch here.

Visualization type: *Example: annotated map / small multiples / linked view*

Goal: *Explain what this sketch tries to reveal.*

Design rationale:

Explain why this visual form supports geographic context, comparison across cities, or simultaneous reading of multiple variables. Mention visual hierarchy, spatial layout, alignment, direct labeling, or other relevant terms from class.

Where this could be useful:

Write where or for whom this visualization would be useful.

Strengths:

- *Strength 1*
- *Strength 2*
- *Strength 3*

Weaknesses:

- *Weakness 1*
- *Weakness 2*

4.3 Sketch 3: Abstract Visualization

Figure 5: Placeholder for Sketch 3 image. Insert the abstract sketch here.

Visualization type: *Example: radial chart / abstract layered composition / symbolic encoding*

Goal: *Explain what this sketch tries to reveal.*

Design rationale:

Explain how the abstract design encodes the data, which visual channels it uses, and why abstraction may help show patterns, contrast, rhythm, or relative change. Mention trade-offs between memorability and precision if relevant.

Where this could be useful:

Write where or for whom this visualization would be useful.

Strengths:

- *Strength 1*
- *Strength 2*
- *Strength 3*

Weaknesses:

- *Weakness 1*
- *Weakness 2*

5 Task 3: Implemented Visualization

For the implementation stage, one of the three visualization concepts was selected and coded as a standalone HTML file with embedded JavaScript.

5.1 Chosen Visualization

Write which sketch you selected and why you selected it for implementation.

5.2 Implementation Details

- **Technology used:** *Example: HTML, CSS, JavaScript, D3.js*
- **File structure:** *Example: single standalone HTML file*
- **Interaction techniques used:** *Example: hover tooltips, highlighting, legends, labels*
- **Visual encoding:** *Brief summary of how variables are represented*

5.3 GitHub Repository

The implementation can be found at the following link:

https://github.com/your-username/your-repository

Replace the placeholder link above with your actual repository link.

5.4 Screenshots of the Final Visualization

Figure 6: Placeholder for screenshot of the implemented visualization.

Figure 7: Placeholder for an additional screenshot, such as interaction state, tooltip, or focused view.

5.5 Implementation Reflection

Write a short reflection about whether the coded version successfully answered the research question, what worked well, and what could still be improved.

6 Discussion

In this section, summarize the main visual patterns you observed after sketching and implementation. For example, mention whether some cities show steady growth while others fluctuate, and whether those changes appear to move together with unemployment, GDP per capita, or inflation. Keep this section interpretive rather than purely descriptive.

7 Conclusion

This homework explored the relationship between urban population change and national socio-economic indicators in Türkiye between 2019 and 2024. The project involved three stages: exploratory sketching, concept generation, and coded implementation. Through these stages, the report aimed to move from raw tabular values toward visual forms that support comparison, trend analysis, and interpretation.

Add 1 short paragraph here summarizing your final takeaway.

Appendix A: Optional Source Image Placeholder

If you want to include the source screenshot that contains the socio-economic indicator values, you may place it here.

Figure 8: Placeholder for the source image containing unemployment, GDP per capita, and inflation values.

Optional note: If the image file is stored locally, replace the placeholder above with a command such as:

```
\includegraphics[width=0.85\textwidth]{tuik_indicators.png}
```

Appendix B: Practical Notes for Replacing Placeholders

- To replace an image placeholder, use:

  ```
  \includegraphics[width=0.9\textwidth]{your-image-file-name}
  ```

- For large hand-drawn sketches, using `[width=0.8\textwidth]` or `[width=0.9\textwidth]` usually works well.
- If you want to place two screenshots side by side, use the `subfigure` environment from the `subcaption` package.
- If your GitHub repository contains a single HTML file, mention that it can be opened directly in a browser.